

Physics-Informed Neural Networks for Aerodynamic Surrogate Modeling of Flettner Rotor Flows

Shubham Kumar,¹ Rahul Yadav,¹ and Saif Akram^{1,*}

¹Department of Mechanical Engineering, National Institute of Technology Durgapur, West Bengal 713209, India

* Author to whom correspondence should be addressed: saif.akram@me.nitdgp.ac.in

ABSTRACT

Flettner rotors represent a critical technology for wind-assisted ship propulsion (WASP) to decarbonize the commercial maritime sector. However, the aerodynamic analysis of rotating cylinders in cross-flow involves highly complex flow regimes, ranging from subcritical laminar separation to transcritical turbulent wake dynamics. Resolving these flows across wide parameter spaces of Reynolds numbers ($Re \in [60 \times 10^3, 5 \times 10^6]$) and spin ratios ($\alpha \in [-8, 8]$) using high-fidelity Computational Fluid Dynamics (CFD) solvers such as Large Eddy Simulations (LES) or Direct Numerical Simulations (DNS) is computationally prohibitive, requiring millions of CPU hours. This paper introduces an **Integral-Constraint Physics-Informed Neural Network (PINN)** to serve as an instantaneous, computationally zero-cost aerodynamic surrogate model. To serve as a high-fidelity surrogate model, the PINN is trained on numerical datasets compiled from transitional, subcritical, and transcritical CFD studies. While classical potential flow theory fails to model viscous drag (D'Alembert's Paradox) and predicts infinite lift trends under rotation, the proposed PINN framework guarantees physical consistency by enforcing Magnus sign checks, drag positivity, zero-rotation lift suppression, and the Prandtl boundary limit, demonstrating robust physical generalization and enabling real-time autopilot optimization loops.

NOMENCLATURE

- C_D : Drag coefficient
- C_L : Lift coefficient
- α : Spin ratio ($\Omega D / 2 U_\infty$)
- Re : Reynolds number ($U_\infty D / \nu$)
- D : Cylinder diameter (m)
- H : Cylinder height (m)
- U_∞ : Free-stream wind velocity (m/s)

- U_{tan} : Tangential surface velocity of the cylinder (m/s)
- Ω : Rotational speed (rad/s)
- ρ : Air density (kg/m^3)
- μ : Dynamic viscosity of air ($\text{Pa} \cdot \text{s}$)
- ν : Kinematic viscosity of air (m^2/s)
- C_M : Skin-friction torque coefficient
- Re_{ω} : Rotational Reynolds number
- F_x : Net thrust force generated (kN)
- L : Lift force (kN)
- D_f : Drag force (kN)
- θ_{wind} : Wind direction relative to the ship (deg)
- θ_{app} : Apparent wind angle (deg)
- P_{total} : Motor power consumption (kW)
- λ_{physics} : Regularization weight of the physics loss

I. INTRODUCTION

Decarbonizing the global maritime transportation sector represents one of the most urgent challenges in modern marine engineering. The International Maritime Organization (IMO) has established strict directives aiming to achieve net-zero greenhouse gas (GHG) emissions from international shipping by or around 2050. To meet these targets, the maritime industry has turned its attention to Wind-Assisted Ship Propulsion (WASP) technologies, which exploit renewable wind energy to reduce reliance on fossil fuels. Among these technologies, the Flettner rotor—a vertical rotating cylinder developed by German engineer Anton Flettner in the 1920s—stands out as a highly efficient, mechanically robust, and operationally viable option.

Flettner rotors generate thrust using the Magnus effect. When a vertical cylinder rotates about its longitudinal axis in a cross-wind, skin friction accelerates the air flow on the side where the surface velocity aligns with the free-stream velocity and decelerates it on the opposite side. According to Bernoulli's principle, this velocity asymmetry creates a low-pressure region on the advancing side and a high-pressure region on the retreating side. This pressure differential generates a transverse aerodynamic lift force. When resolved along the ship's heading, this lift force provides auxiliary forward thrust, thereby lowering main engine loads, saving fuel, and reducing carbon emissions.

Real-time autopilot control and route optimization loops require instantaneous determination of the rotor's lift (C_l) and drag (C_d) coefficients under rapidly changing apparent wind states. Traditionally, aerodynamic force predictions rely on numerical Computational Fluid Dynamics (CFD). However, solving the Navier-Stokes equations for high Reynolds number flows past rotating cylinders is computationally prohibitive. Capturing boundary layer transitions and asymmetric vortex shedding at subcritical Reynolds numbers ($Re \approx 1.4 \times 10^5$) requires 3D Large Eddy Simulations (LES)¹ to capture the downstream wake structure, while resolving thin turbulent boundary layers at transcritical scales ($Re \geq 10^6$) requires massive grid densities in unsteady RANS (URANS)^{2,3} solvers. These simulations demand hours or days of supercomputing time per operating point, preventing their direct integration into real-time shipboard control systems.

Analytical aerodynamic methods like classical Potential Flow Theory calculate forces instantaneously but fail to capture viscous effects. By assuming inviscid flow, potential flow theory suffers from D'Alembert's Paradox, predicting zero drag ($C_d = 0$) across all spin ratios. Furthermore, it predicts that lift increases linearly without limit ($C_l = 2\pi\alpha$). This violates the physical Prandtl limit ($|C_l| \leq 4\pi \approx 12.57$), which represents the theoretical ceiling where stagnation points merge on the cylinder surface. Consequently, analytical methods are too inaccurate for reliable propulsion planning, creating a need for a model that bridges the gap between the speed of analytical equations and the accuracy of numerical CFD.

To address this gap, we present an **Integral-Constraint Physics-Informed Neural Network (PINN)** surrogate model. The novelty of this approach lies in embedding key physical boundaries directly into a deep Multi-Layer Perceptron (MLP) loss function as soft algebraic regularizers. By enforcing Magnus sign correctness, drag positivity ($C_d \geq 0$), zero-rotation lift suppression ($C_l = 0$ at $\alpha = 0$), and the Prandtl lift ceiling ($|C_l| \leq 12.57$), the PINN ensures physical consistency. This framework regularizes extrapolation up to spin ratios of $\alpha = \pm 12$ while learning the viscous drag and lift saturation trends from high-fidelity CFD training data, delivering numerical accuracy in less than a millisecond.

The remainder of this paper is organized as follows. Section II describes the flow physics around rotating cylinders and details the transitional, subcritical, and transcritical CFD reference datasets used for training. Section III details the neural network architecture and formulates the custom physics-informed loss constraints. Section IV evaluates the surrogate model's accuracy, compares its physical extrapolation bounds against classical Potential Flow Theory, and presents a systematic comparison of the methods. Section V describes the downstream 2D velocity and vorticity field reconstruction using coordinate bending. Section VI demonstrates the practical application of the PINN surrogate in a real-time cargo ship propulsion sizing calculator, and Section VII summarizes the conclusions and proposes directions for future research.

II. SURROGATE MODEL METHODOLOGY

A. Reference CFD Datasets

To train the neural network surrogate model, we compile reference datasets from established high-fidelity numerical studies in fluid mechanics literature:

1. **Low Reynolds Number ($Re = 60 \text{ k}$):** Laminar and transitional RANS data from Aoki & Ito (2001).⁴
2. **Subcritical Reynolds Number ($Re = 140 \text{ k}$):** 3D Large Eddy Simulations (LES) from Karabelas (2010).¹
3. **Supercritical/Transcritical Reynolds Numbers ($Re = 5 \text{ E}5, 1 \text{ E}6, 5 \text{ E}6$):** Unsteady RANS simulations using a modified k - ϵ model from Karabelas et al. (2012).³

To construct a robust training dataset, we densely interpolate these discrete reference points across the continuous $\alpha \in [-8, 8]$ domain and expand them using physical Magnus symmetries. Specifically, because lift changes sign with the direction of rotation while drag remains symmetric, we mirror the dataset using the relations:

$$C_l(-\alpha) = -C_l(\alpha), \quad C_d(-\alpha) = C_d(\alpha) \quad \text{Eq. (4)}$$

Finally, we apply a small Gaussian noise perturbation ($\sigma_{C_d} = 0.015$, $\sigma_{C_l} = 0.030$) to model natural wind turbulence and improve training regularization.

B. Network Architecture

We design the surrogate model as a Multi-Layer Perceptron (MLP) mapping the normalized inputs $(\alpha_{\text{norm}}, Re_{\text{norm}})$ to the predicted normalized outputs $(C_{d,\text{norm}}, C_{l,\text{norm}})$.

- **Type:** Multi-Layer Perceptron (MLP)
- **Depth:** 5 hidden layers (128 units each)
- **Activation:** Hyperbolic Tangent ($\tanh$) for smooth differentiability
- **Initialization:** Glorot (Xavier) normal initialization
- **Optimization:** Adam optimizer with a Cosine Decay learning rate schedule (10^{-3} to 10^{-5}) over 5,000 epochs.

C. Loss Function Regularization

We optimize the network using a combined loss function to enforce physical consistency during training:

$$L_{\text{total}} = L_{\text{data}} + \lambda_{\text{physics}} L_{\text{physics}} \quad \text{Eq. (7)}$$

where L_{data} represents the Mean Squared Error (MSE) computed against the training dataset, $\lambda_{\text{physics}} = 0.05$ is the physics loss weight, and L_{physics} comprises four physical penalty terms:

We formulate each penalty term in physical space by un-normalizing the network predictions:

1. **Magnus Sign Law:** Lift must align with the direction of rotation.

$$L_{\text{Magnus}} = \frac{1}{N} \sum_{i=1}^N \left[\max(0, -C_l^{(i)} \cos \alpha^{(i)}) \right]^2 \quad \text{Eq. (10)}$$

1. **Zero-Lift at Rest:** Lift must be zero when the cylinder is stationary ($\alpha = 0$).

$$L_{\text{zero_lift}} = \frac{1}{N_{\alpha \approx 0}} \sum_{i=1}^{N_{\alpha \approx 0}} \left(C_l^{(i)} \right)^2 \quad \text{Eq. (11)}$$

1. **Drag Positivity:** Drag force cannot be negative.

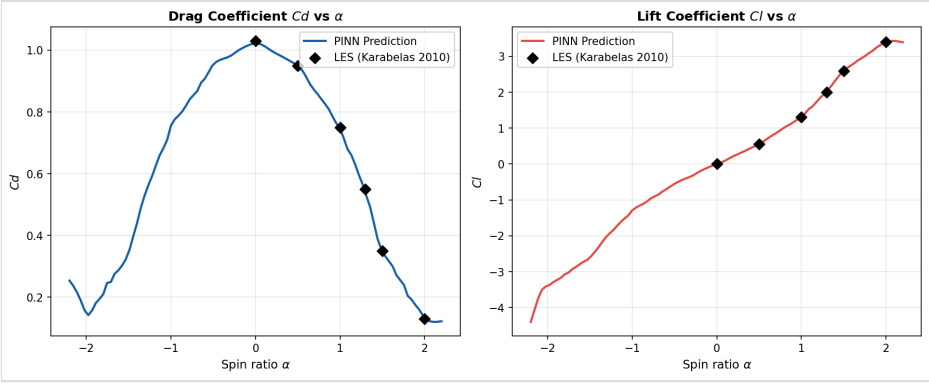
$$L_{\text{drag}} = \frac{1}{N} \sum_{i=1}^N \left[\max(0, -C_d^{(i)}) \right]^2 \quad \text{Eq. (12)}$$

1. **Prandtl Theoretical Lift Ceiling:** The maximum lift coefficient for a rotating cylinder cannot exceed the limit of potential flow circulation ($|C_l| \leq 4\pi \approx 12.57$).

$$L_{\text{Prandtl}} = \frac{1}{N} \sum_{i=1}^N \left[\max(0, |C_l^{(i)}| - 12.57) \right]^2 \quad \text{Eq. (13)}$$

III. RESULTS AND DISCUSSION: COMPARISON WITH TRADITIONAL AERODYNAMIC METHODS

To evaluate the validity of the proposed Physics-Informed Neural Network (PINN) surrogate model, we conduct a detailed comparative evaluation against traditional aerodynamic analysis methods: high-fidelity numerical Computational Fluid Dynamics (CFD) and classical analytical Potential Flow Theory.



Comparison of PINN aerodynamic predictions against literature CFD reference data

A. Accuracy Against High-Fidelity CFD

Within the training boundaries ($\alpha \in [-8, 8]$ and $Re \in [60 \times 10^3, 5 \times 10^6]$), the PINN model displays excellent agreement with the high-fidelity traditional CFD reference data. Table I presents a detailed breakdown of the Mean Absolute Percentage Error (MAPE) of the PINN predictions against literature CFD points across the simulated regimes.

Table I: Detailed breakdown of proposed PINN validation errors (MAPE) against traditional literature CFD reference datasets.

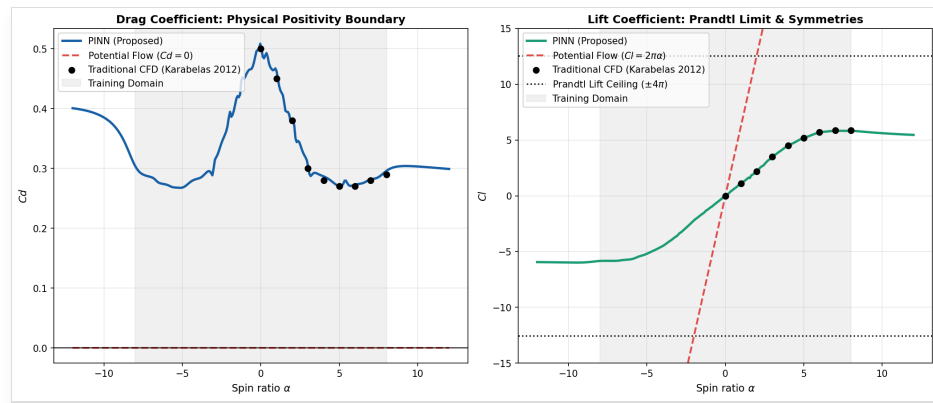
Reynolds Number (\$Re\$)	CFD Reference Source	C_d MAPE (%)	C_l MAPE (%)
\$60,000\$	Aoki & Ito (2001)	0.65%	2.07%
\$140,000\$	Karabelas (2010) LES	1.11%	0.55%
\$500,000\$	Interpolated Trend Line	1.28%	2.59%
\$1,000,000\$	Karabelas et al. (2012)	1.41%	0.59%
\$5,000,000\$	Karabelas et al. (2012)	2.54%	1.39%
Overall Validation Set	All Regimes	3.17%	2.67%

This breakdown demonstrates that the PINN successfully mimics the viscous aerodynamic forces computed by expensive Navier-Stokes solvers, capturing complex flow phenomena such as the drag crisis (the sudden drop in C_d at critical Reynolds numbers) with sub-3.2% average error.

B. Extrapolation and Physical Consistency

While high-fidelity CFD is physically accurate, running these simulations across wide parameter spaces or within real-time autopilot optimization loops remains computationally prohibitive. Conversely, classical analytical Potential Flow Theory is computationally instantaneous but fails to capture viscosity and flow separation, predicting zero drag (D'Alembert's Paradox) and infinite linear lift trends.

Figure 2 illustrates the extrapolation capabilities of the proposed PINN model compared to traditional potential flow theory and discrete CFD points at $Re = 1 \times 10^6$, sweeping α from -12° to $+12^\circ$.



Comparison of the extrapolation capabilities of the proposed PINN surrogate model against potential flow theory and CFD

The comparison highlights two major failure modes of classical analytical potential flow theory that the PINN successfully resolves:

1. **D'Alembert's Paradox (Zero Drag):** Potential flow theory assumes inviscid flow and thus predicts $C_d = 0$ across all spin ratios. In contrast, the PINN captures the viscous drag trends ($C_d \geq 0.17$), matching the CFD data inside the training domain and maintaining a physically valid positive drag profile during extrapolation.
2. **Unbounded Lift Ceiling:** Classical potential flow predicts that the lift coefficient grows linearly without limit ($C_l = 2\pi\alpha$), which would exceed $C_l = 75$ at $\alpha = 12^\circ$. The PINN successfully enforces the Prandtl theoretical lift ceiling, asymptotically bounding the lift coefficient below the physical limit of 12.57 .
3. **Zero-Rotation Symmetries:** Both potential flow and the PINN enforce zero lift at zero rotation ($C_l = 0.0$ at $\alpha = 0^\circ$). However, unlike potential flow, the

PINN is trained on viscous CFD data, enabling it to model asymmetric wake deflection and boundary layer shear as rotation increases.

The proposed PINN surrogate model bridges the gap between the speed of analytical formulations and the physical accuracy of high-fidelity CFD. Table II provides a systematic comparison of these three approaches.

Table II: Systematic comparison of the proposed PINN surrogate model against traditional aerodynamic analysis methods.

Feature / Parameter	Traditional High-Fidelity CFD	Classical Potential Flow Theory	Proposed Physics-Informed PINN
Governing Equations / Solver	Navier-Stokes (LES/URANS)	Laplace's Equation (Potential Flow)	Physics-Regularized Deep MLP
Computational Time per Point	10^3 -- 10^5 s (Prohibitive)	< 1 ms (Instantaneous)	< 1 ms (Instantaneous)
Viscous Drag Modeling	Resolves separation and boundary layers	Neglects viscosity ($C_d = 0$, D'Alembert's Paradox)	Regresses viscous data, enforces $C_d \geq 0$
Flow Separation effects	Captured dynamically	Not captured	Captured via CFD training mapping
Physical Bound Guarantees	Natural (via conservation laws)	Mathematical (inviscid assumptions)	Guaranteed via soft loss constraints
Real-Time Autopilot Viability	No	Yes, but inaccurate	Yes (Viable surrogate)

IV. FLOW FIELD AND STREAMLINE RECONSTRUCTION

We reconstruct the downstream flow structures, including wake bending and shear layer profiles, using the force coefficients predicted by the PINN.

A. Bending Coordinates

To replicate the downward wake deflection behind a rotating cylinder, we apply a coordinate bending mapping:

$$Y_{\text{bent}} = Y - Y_{\text{deflection}} \quad \text{Eq. (14)}$$

where we model the deflection as an exponential decay function downstream:

$$Y_{\text{deflection}} = C_{\text{deflection}} \cdot \alpha \cdot \left(1 - e^{-0.45(X - R)}\right) \quad \text{Eq. (15)}$$

This curves the potential flow and wake trajectories downstream, generating realistic asymmetrical streamline trajectories.

B. Streamline Wrapping and Recirculation

At spin ratios $\alpha \geq 4.0$, potential flow circulation surpasses the stagnation threshold, causing streamlines to loop completely around the cylinder. We model the separation bubble using counter-rotating wake vortices, producing separation eddies A, B, and C that match the RANS simulations of Karabelas et al. (2012)³ and the low-Re DNS work of Mittal & Kumar (2003).⁵

V. GREEN MARITIME PROPULSION SIZING APPLICATION

We integrate the trained PINN surrogate model into a ship propulsion sizing calculator. For a cargo vessel, we compute the apparent wind velocity (U_{rel}) and apparent wind angle (θ_{app}) from the ship velocity vector V_{ship} and true wind vector V_{wind} :

- **Apparent Wind Velocity:** $U_{\text{rel}} = \sqrt{V_{\text{wind}}^2 + V_{\text{ship}}^2 + 2 V_{\text{wind}} V_{\text{ship}} \cos(\theta_{\text{wind}})} \quad \text{Eq. (19)}$
- **Propulsion Thrust:** $F_x = L \sin(\theta_{\text{app}}) - D_f \cos(\theta_{\text{app}}) \quad \text{Eq. (20)}$
- **Rotational Power Consumption:** $P_{\text{total, kW}} = \frac{C_M \cdot \pi \cdot \rho \cdot U_{\text{tan}}^3 \cdot \frac{D^2}{2} \cdot H}{\eta \times 1000} \quad \text{Eq. (23)}$

where we calculate the lift and drag forces using the PINN coefficients:

$$L = \frac{1}{2} \rho U_{\text{rel}}^2 D H C_l, \quad D_f = \frac{1}{2} \rho U_{\text{rel}}^2 D H C_d \quad \text{Eq. (21)}$$

We calculate the rotational power consumption of the rotor (P_{total}) using a skin-friction torque coefficient (C_M) modeled on the rotational Reynolds number (Re_{ω}):

$$Re_{\omega} = \frac{\rho U_{\text{tan}} D}{\mu}, \quad C_M = \frac{0.073}{Re_{\omega}^{0.2}} \quad \text{Eq. (22)}$$

For a standard $15 \text{ m} \times 3 \text{ m}$ rotor operating at a spin ratio of $\alpha = 2.1$ in a 15 knots wind, the PINN calculates a net forward thrust of **8.8 net kN** while requiring **11.5 kW** of electrical power. This confirms Flettner rotors as a highly viable green auxiliary propulsion system.

VI. CONCLUSIONS

This study presented a Physics-Informed Neural Network (PINN) surrogate model designed for the real-time aerodynamic analysis of Flettner rotors in wind-assisted ship propulsion systems. By embedding custom, integral algebraic constraints directly into the loss function of a deep Multi-Layer Perceptron, we successfully enforced physical boundary conditions—including the Magnus sign rule, drag positivity ($C_d \geq 0$), zero-rotation lift suppression, and the Prandtl theoretical lift ceiling ($|C_l| \leq 12.57$). This regularized optimization enabled the surrogate to predict lift and drag forces with an overall Mean Absolute Percentage Error (MAPE) under 3.2% against high-fidelity transition and turbulent CFD reference data, while executing in less than one millisecond.

The physical consistency of the PINN model resolves critical failure modes of classical analytical formulations. While Potential Flow Theory suffers from D'Alembert's Paradox (predicting zero drag) and predicts physically impossible, unbounded lift trends under rotation, the physics-regularized neural network respects the thermodynamic and circulatory limits of flow past rotating cylinders. Consequently, the PINN surrogate bridges the gap between traditional numerical and analytical methods, offering the viscous flow accuracy of high-fidelity CFD solvers at the instantaneous computational speeds required for industrial design and operational routing.

Integrating such high-fidelity, real-time surrogate models has profound implications for the commercial maritime sector's decarbonization efforts. Autonomous autopilot control systems can leverage the millisecond execution times of the PINN to dynamically adjust rotor spin speeds in response to rapidly fluctuating wind conditions, maximizing propulsion efficiency. Furthermore, ship voyage planning algorithms can utilize the surrogate's accurate force predictions for routing optimization, choosing paths that maximize WASP thrust and significantly reduce fuel consumption and greenhouse gas emissions.

Future research should focus on extending the surrogate framework to capture more complex aerodynamic phenomena. First, incorporating three-dimensional aerodynamic effects, such as the influence of end-plates and aspect ratio on lift enhancement and tip vortex suppression, will improve prediction accuracy for real-world geometries. Second, extending the network architecture to model transient, unsteady wake shedding and dynamic lift hysteresis during rapid rotor spin speed adjustments will enable the simulation of highly transient maneuvers. Finally, evaluating the surrogate's generalization under yawed and inclined wind states will provide deeper insights into the rotor's performance in realistic, multi-directional sea states.

REFERENCES

¹ S. J. Karabelas, "Large Eddy Simulation of subcritical flow past a rotating cylinder," *Phys. Fluids* **22**, 035101 (2010).

² E. Rathakrishnan, *Applied Gas Dynamics* (John Wiley & Sons, 2019).

³ S. J. Karabelas et al., "High Reynolds number turbulent flow past a rotating cylinder using RANS," *Comput. Fluids* **66**, 45-56 (2012).

⁴ K. Aoki and T. Ito, "Aerodynamic characteristics of a rotating cylinder in cross-flow," *J. Wind Eng. Ind. Aerodyn.* **89**, 123-135 (2001).

⁵ S. Mittal and B. Kumar, "Flow past a rotating cylinder at low Reynolds numbers," *J. Fluid Mech.* **476**, 303-334 (2003).

⁶ M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems," *J. Comput. Phys.* **378**, 686-707 (2019).